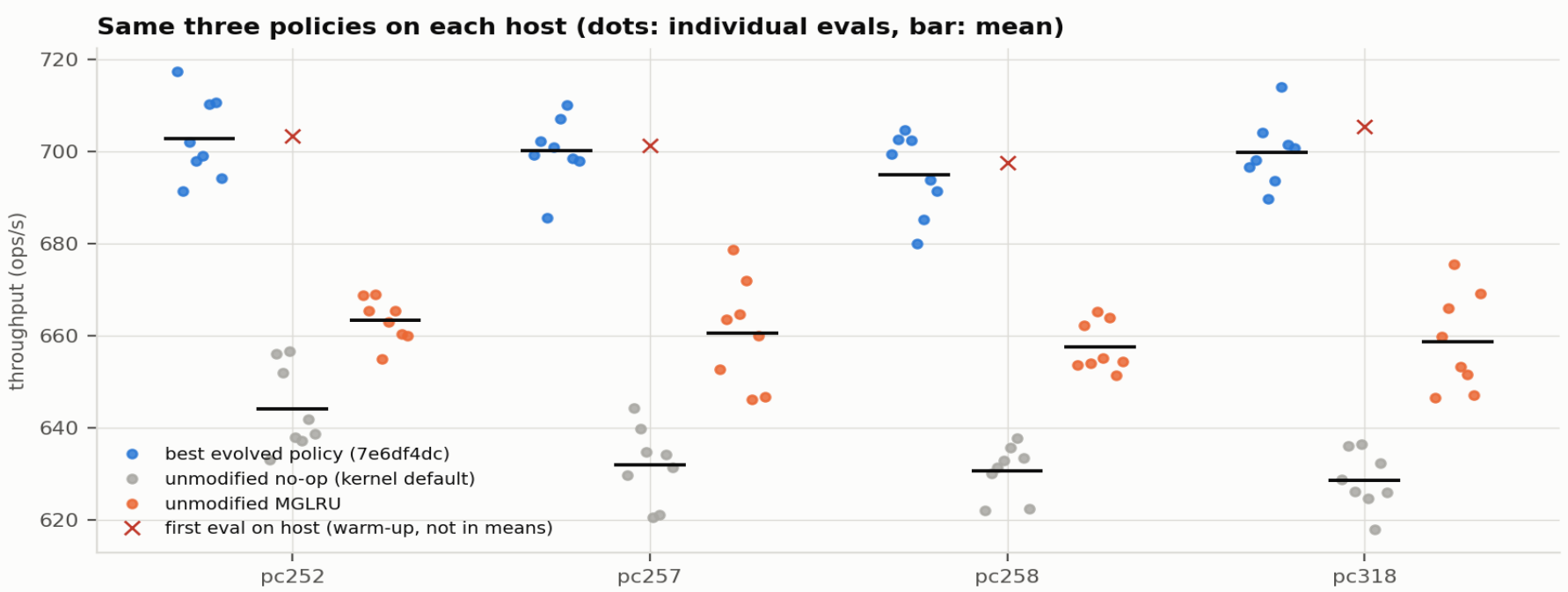
Does the 712 ops/s policy hold up on the four new hosts?

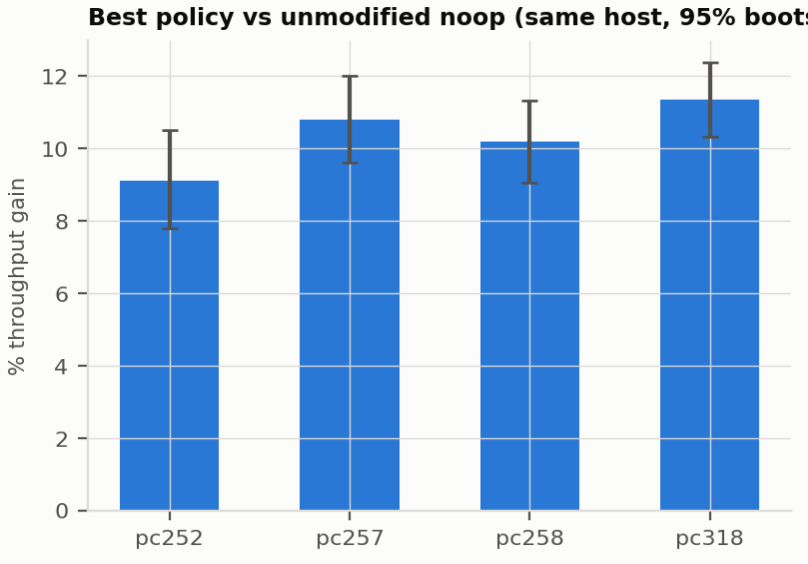
The best policy from the earlier 4-host run (7e6df4dc, region-CLOCK, 712.2 ops/s on pc309, evolved on a different set of hosts: pc306, pc309, pc319, pc320) was run unchanged on the four current hosts (pc252, pc257, pc258, pc318), interleaved with two unmodified controls (the no-op kernel default and MGLRU) so that host speed can be separated from policy effect. 8 evaluations per policy per host, order rotated each round, one warm-up eval per host (not in the means). Hosts ran in parallel, one eval at a time each.

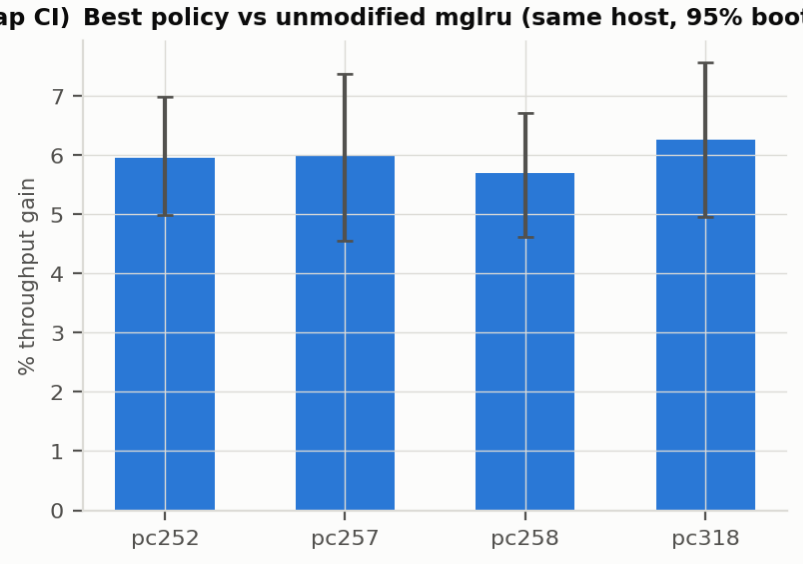
host	best policy mean ± CI95 (sd, cv)	no-op mean ± CI95	MGLRU mean ± CI95	warm-up eval	speed index (no-op)	gain vs no-op [95% CI]	gain vs MGLRU [95% CI]
pc252	702.8 ± 7.5 (sd 9.0, cv 1.3%)	644.1 ± 7.8	663.3 ± 4.0	703.2	1.016	+9.1% [+7.8, +10.5]	+6.0% [+5.0, +7.0]
pc257	700.1 ± 6.1 (sd 7.3, cv 1.0%)	631.9 ± 6.9	660.5 ± 9.7	701.2	0.997	+10.8% [+9.6, +12.0]	+6.0% [+4.6, +7.4]
pc258	694.9 ± 7.5 (sd 9.0, cv 1.3%)	630.6 ± 4.8	657.5 ± 4.5	697.5	0.995	+10.2% [+9.0, +11.3]	+5.7% [+4.6, +6.7]
pc318	699.7 ± 6.1 (sd 7.3, cv 1.0%)	628.5 ± 5.2	658.6 ± 9.0	705.3	0.992	+11.3% [+10.3, +12.4]	+6.3% [+4.9, +7.6]

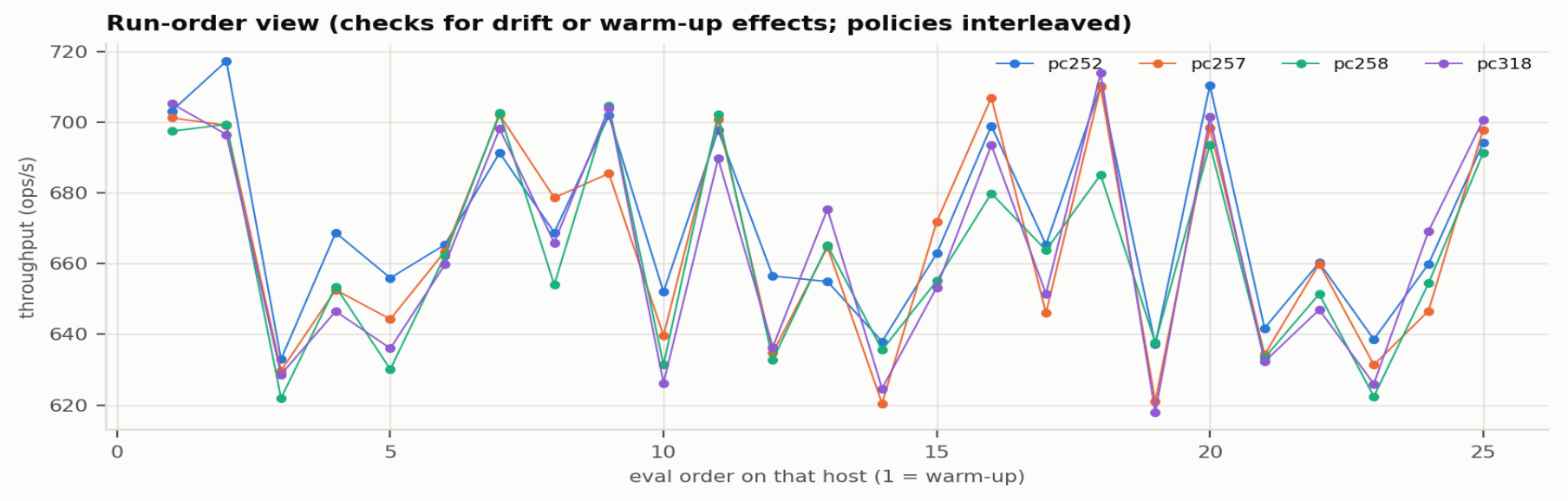
Do hosts differ in how much the policy helps? Pairwise difference in the best-vs-no-op gain (log-ratio, % points) with 95% bootstrap CI: pc252 vs pc257: -1.5 [-3.2, +0.2]; pc252 vs pc258: -1.0 [-2.6, +0.6]; pc252 vs pc318: -2.0 [-3.6, -0.4] *; pc257 vs pc258: +0.5 [-0.9, +2.1]; pc257 vs pc318: -0.5 [-1.9, +0.9]; pc258 vs pc318: -1.0 [-2.5, +0.4]. * = CI excludes 0.

Drift over the experiment (change in normalised throughput from first to last eval, linear fit): pc252 -1.2%, pc257 -1.0%, pc258 -0.8%, pc318 +0.1%.


Same three policies on each host (dots: individual evals, bar: mean)
throughput (ops/s)
720
700
680
660
640
620
best evolved policy (7e6df4dc)
unmodified no-op (kernel default)
unmodified MGLRU
first eval on host (warm-up, not in means)
pc252
pc257
pc258
pc318


Best policy vs unmodified noop (same host, 95% bootstrap CI)
% throughput gain
12
10
8
6
4
2
0
pc252
pc257
pc258
pc318


Best policy vs unmodified mglru (same host, 95% bootstrap CI)
% throughput gain
7
6
5
4
3
2
1
0
pc252
pc257
pc258
pc318



Per host and policy

host	policy	n	mean	sd	min	max	MiB read (mean)	refaults (mean)
pc252	best	8	702.8	9.0	691.4	717.3	194	23855
pc252	noop	8	644.1	9.3	633.0	656.5	180	20563
pc252	mglru	8	663.3	4.8	654.9	668.8	192	23589
pc257	best	8	700.1	7.3	685.5	710.0	193	23630
pc257	noop	8	631.9	8.3	620.5	644.3	183	21083
pc257	mglru	8	660.5	11.6	646.0	678.7	193	23859
pc258	best	8	694.9	9.0	679.9	704.7	190	23088
pc258	noop	8	630.6	5.7	622.0	637.7	178	20024
pc258	mglru	8	657.5	5.4	651.4	665.1	189	22974
pc318	best	8	699.7	7.3	689.7	713.9	194	23968
pc318	noop	8	628.5	6.2	617.9	636.4	180	20601
pc318	mglru	8	658.6	10.7	646.5	675.4	195	24394